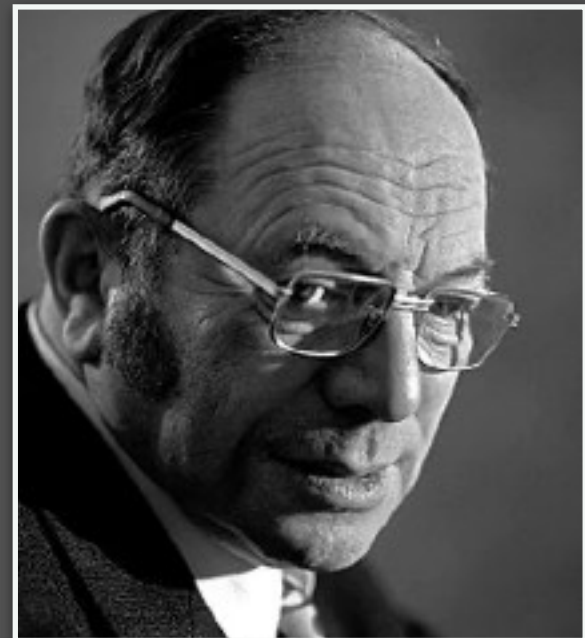


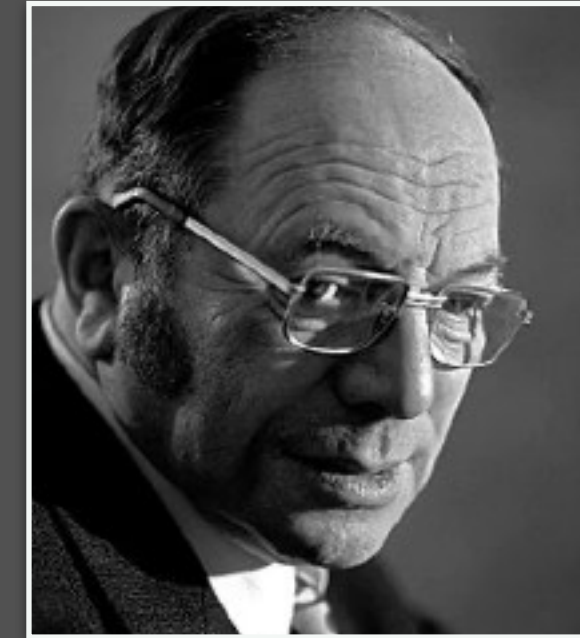
Last Time



Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i=1}^n \sum_{j=1}^m C_{i,j} \pi_{i,j}$$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$

- **Feasible set:** couplings with marginals α, β
- **Interpretation:** average “work” to move mass $\mathbb{E}_{\pi}(c(X, Y))$
- **Existence:** guaranteed to have a solution, but not necessarily unique

Today

- From **Kantorovich** to **Wasserstein**: definition and special cases
- **Metric Properties**: axioms, Wasserstein space and its geometry
- **Duality**: an alternative view of the problem